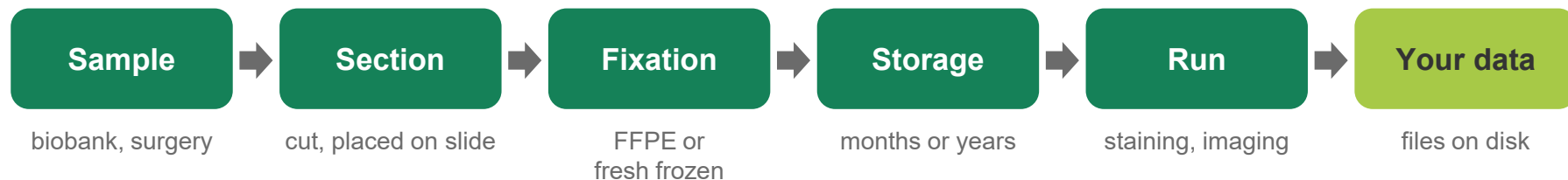


Imaging-based spatial data: preprocessing, QC and normalization

Christophe Avenel, NBIS
ELIXIR SCO Spatial Omics Data Analysis, Uppsala, 6 October 2026

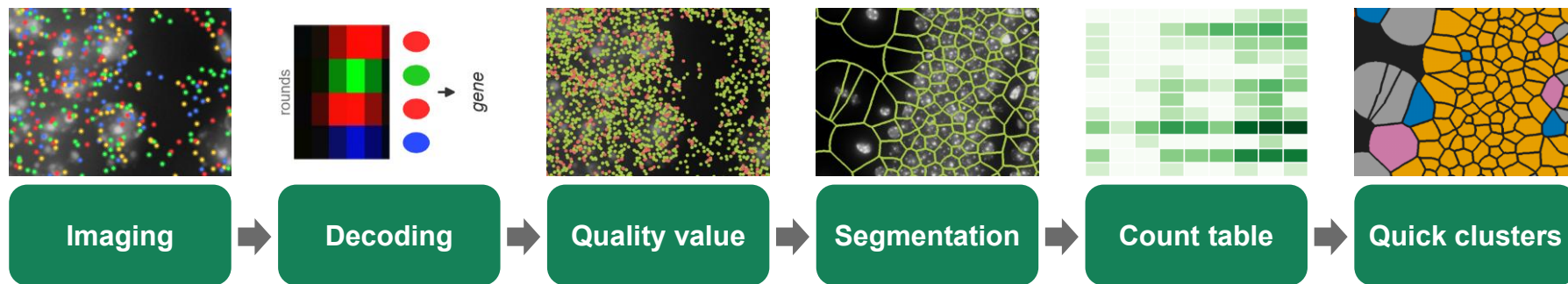
A lot happens before your data



Each step can lose RNA or damage the tissue. **QC can find damage, but it cannot undo it.**

What the instrument already did

Xenium Onboard Analysis (XOA) runs on the instrument



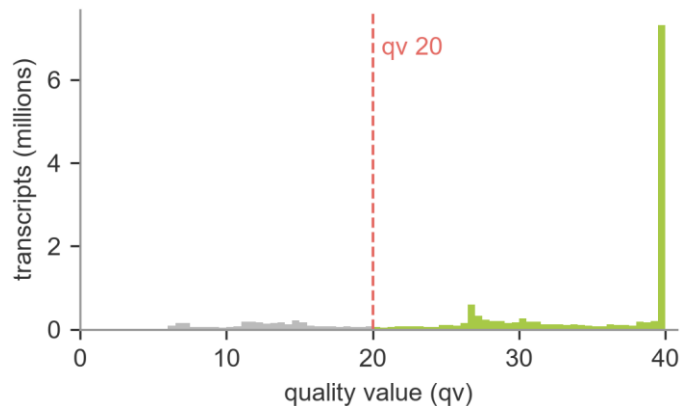
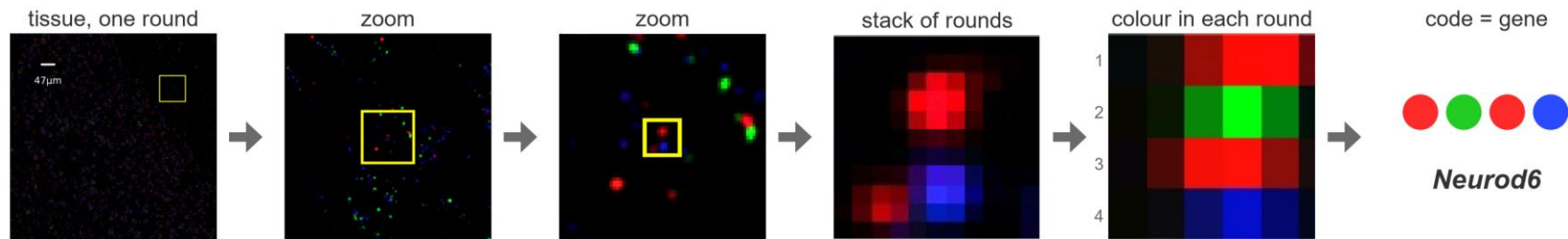
You can redo

- **Xenium Ranger**: relabel, resegment, import-segmentation
- **MERSCOPE**: VPT (Vizgen post-processing tool)
- **CosMx**: AtoMx

Software version is a batch variable

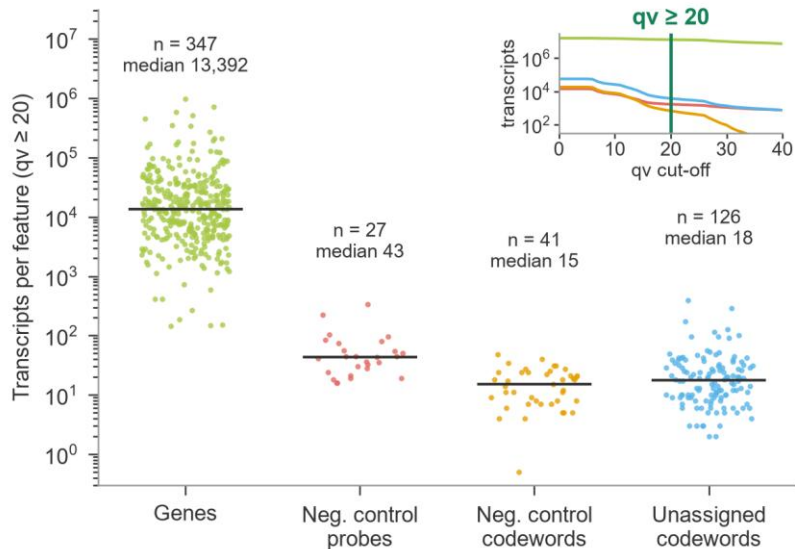
- 10x: mixing XOA 4.0 data with older versions can create batch effects
- Record versions and settings for every sample
- Our data: XOA 1.4, nucleus + 15 μ m expansion

Decoding and the quality value



- **qv**: how sure the match is. qv 20 = about 1 error in 100
- Only qv ≥ 20 goes into the count table. `transcripts.parquet` keeps all
- **Our data**: 82% of gene transcripts pass

Controls show the background



Control	Measures
Negative control probe	non-specific binding + decoding errors
Negative control codeword	decoding errors
Unassigned codeword	decoding errors (unused codes)
Genomic control (5K)	signal from DNA

5 genes are not above the highest negative probe:
Tnf, Cd69, Lcn2, Slc6a3, Nme8

No control detects a probe that binds the wrong gene.

Is my run good?

Metric (metrics_summary.csv)	Our data	10x warning
Gene transcripts with $qv \geq 20$	82%	low: check images
Transcripts per 100 μm^2 of cells	63.5	< 10
Transcripts inside cells	95%	< 50%
Empty cells	2.3%	> 10%
Adjusted negative control probe rate	0.18%	high: background
Median transcripts / genes per cell	181 / 66	panel dependent

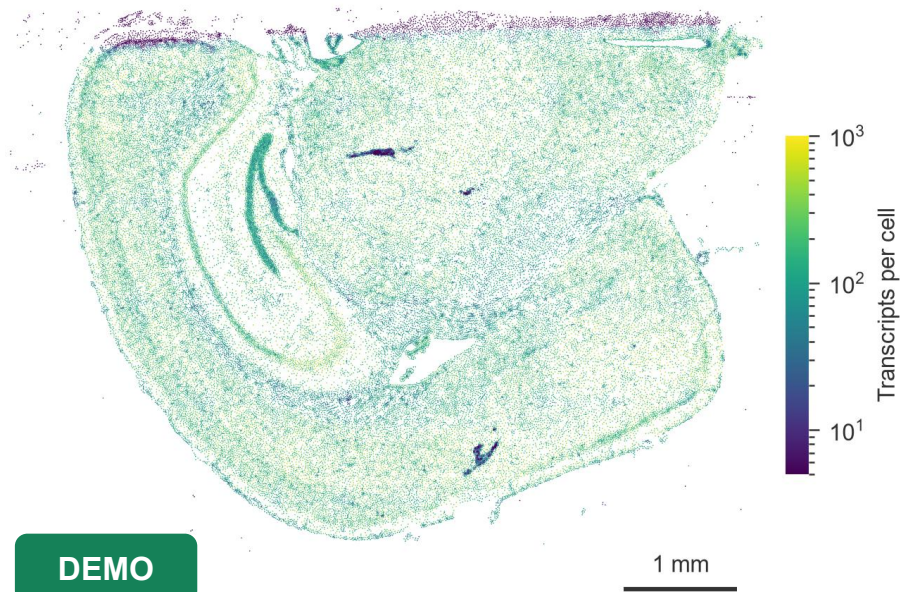
95% in cells?

High because of the 15 μm expansion.
Only 29% overlap a nucleus.

Compare with other runs

Spatial Touchstone: standard metrics and reference ranges, three platforms.

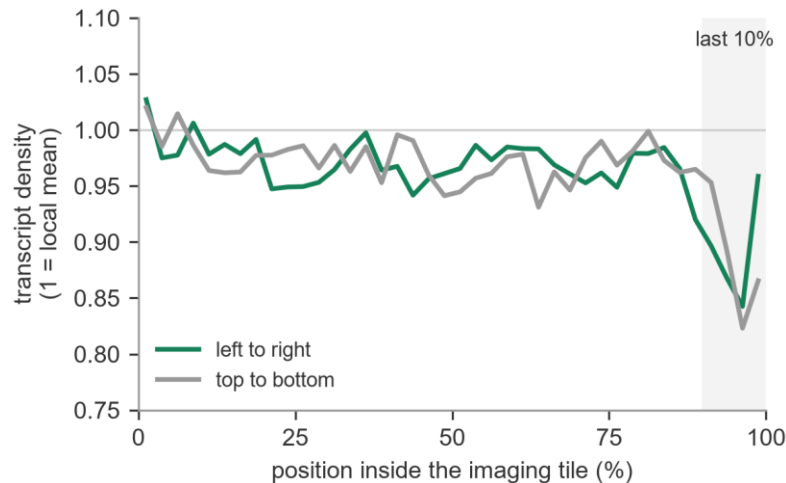
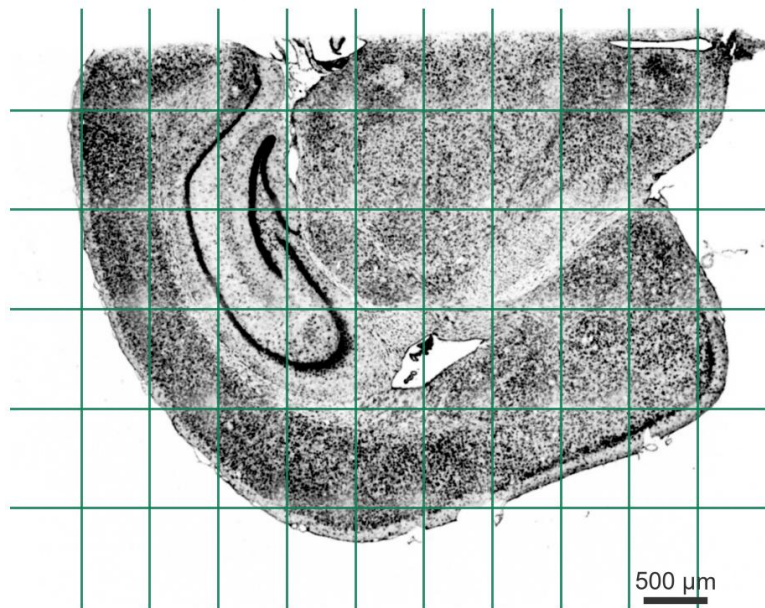
What can go wrong in the tissue



Problem	How it looks
Detached or lost tissue	empty patches
Fold	doubled nuclei, high counts
Edge effect	low-count band
Imaging tiles (FOV)	grid pattern
Autofluorescence	high control codewords
Our data: a low-count band along the top edge.	

Edges of the imaging tiles

transcripts, with the 66 imaging tiles



Last 10% of each tile:

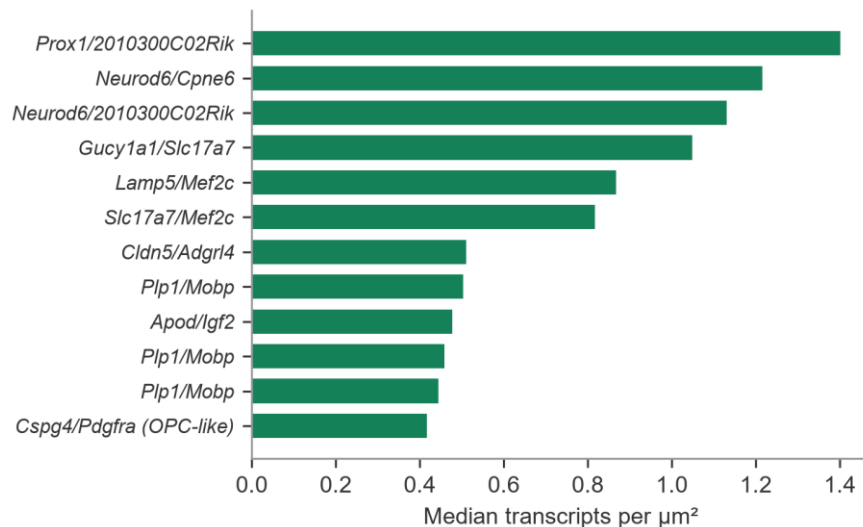
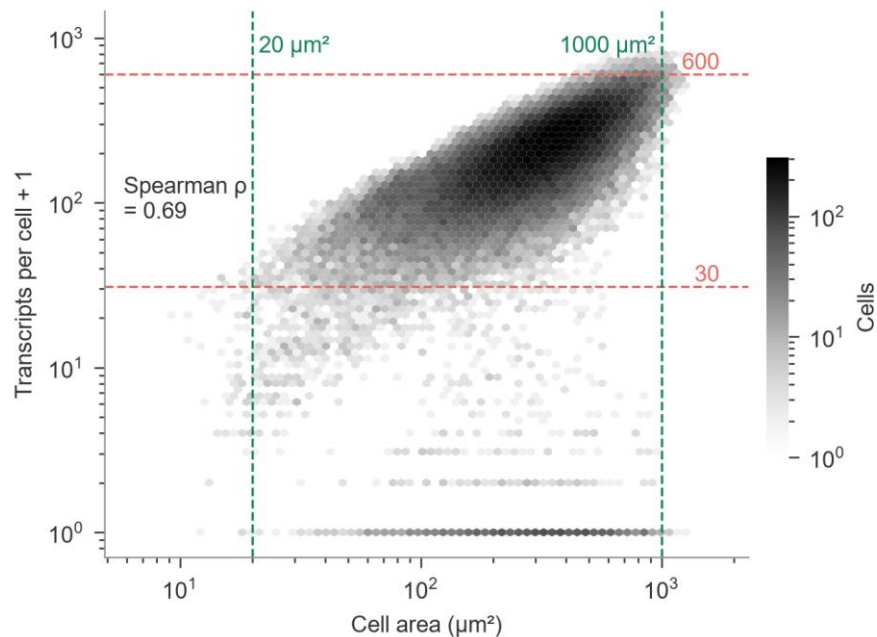
18% of cells, about 15% fewer counts

What we measure per cell

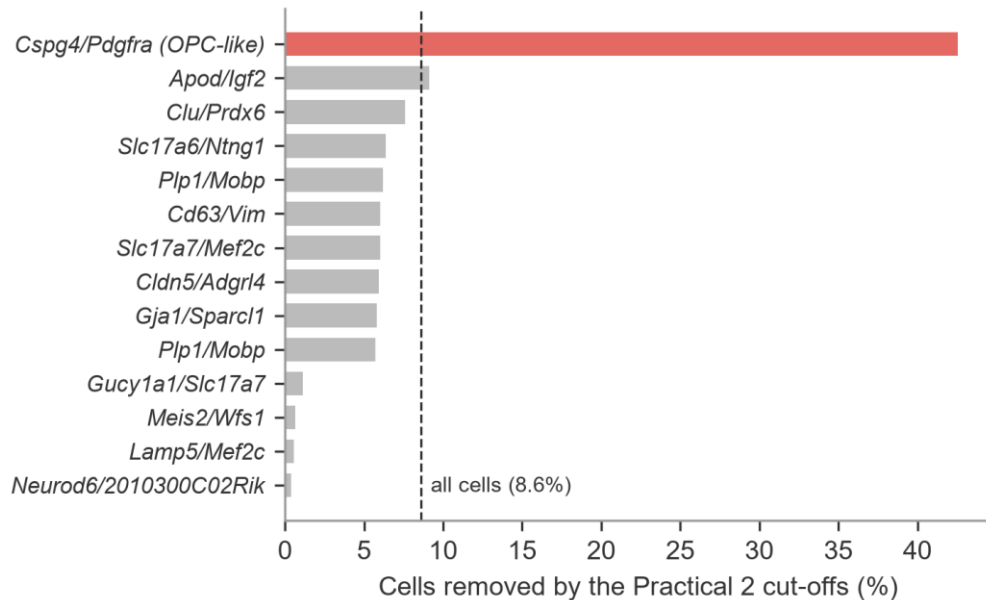
Metric	Source	What it shows
Transcripts, genes	transcript_counts; genes counted from the matrix	how much RNA was detected
Cell and nucleus area	cell_area, nucleus_area	segmentation size
Counts per μm^2	counts / cell_area	signal density
Nucleus / cell ratio	nucleus_area / cell_area	how much was expanded
Control counts	control_probe_counts, ...	background in the cell
Distance to tile border	(CosMx)	cut cells at the border

Most panels have no mitochondrial genes, so there is no % mito metric.

Counts depend on size and on cell type



Cut-offs remove some cell types more than others



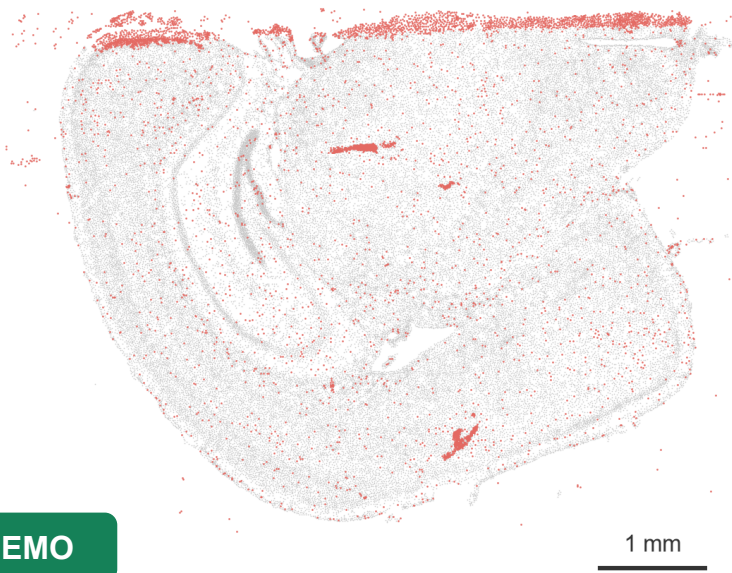
Same cut-offs, very different losses

- 43% of OPC-like cells are removed, under 1% of some neurons
- Low counts can be biology
- Check what you remove, per cell type

Always plot what you remove, in space

● kept (56,899 cells)

● removed (5,369 cells, 8.6%)



DEMO

44% of removed cells are in the top band

- Remove or flag the bad region first, then filter cells
- Is the region damaged, or real tissue with few cells?

Thresholds in practice

Platform and panel	Minimum counts
Xenium onboard analysis	5
Xenium, Marco Salas 2025, Bilous 2026	10
MERSCOPE	10
CosMx 1K / 6K / WTX	20 / 50 / 100

More genes, more counts

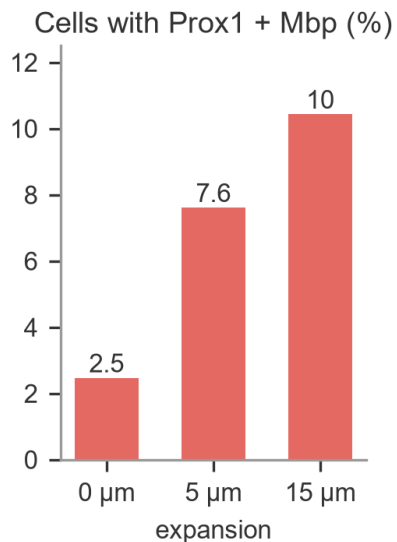
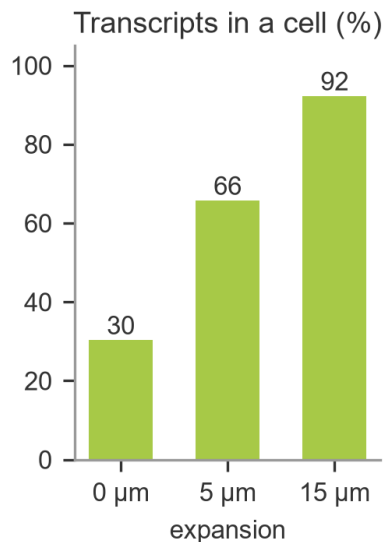
- The threshold should grow with the panel size
- Many analyses use one fixed value for every panel
- **Rule of thumb:**
remove at most 5-10% of cells.
Practical 2: 8.6%

Alternatives to fixed cut-offs

Approach	Idea	Where
Fixed cut-off	same value for all cells	vendor defaults, most analyses
Outliers by MAD	far from the median, per sample, log scale	OSTA book, scuttle
Combined score	area + density + shape + controls → one score	SpaceTrooper (R)

- Keep small cells if their density and controls look normal
- MAD-based QC is standard in scRNA-seq, less used in spatial data so far

Contamination is also QC

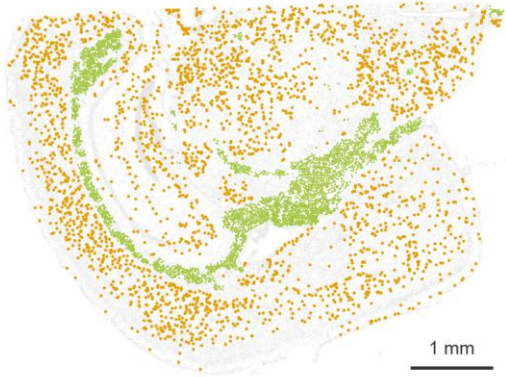


- Transcripts from neighbours end up in the wrong cell
- Like doublets in scRNA-seq
- Options: nucleus-only counts, Proseg, SPLIT, ResolVI
- Details: segmentation lecture

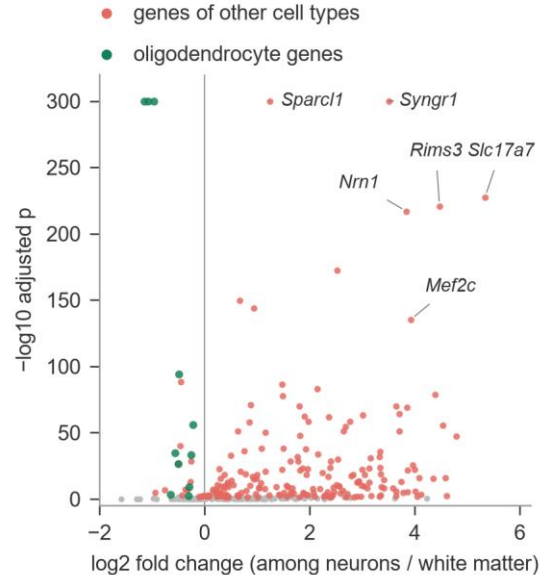
Contamination looks like differential expression

Oligodendrocytes

- in white matter (4,038)
- among neurons (2,833)

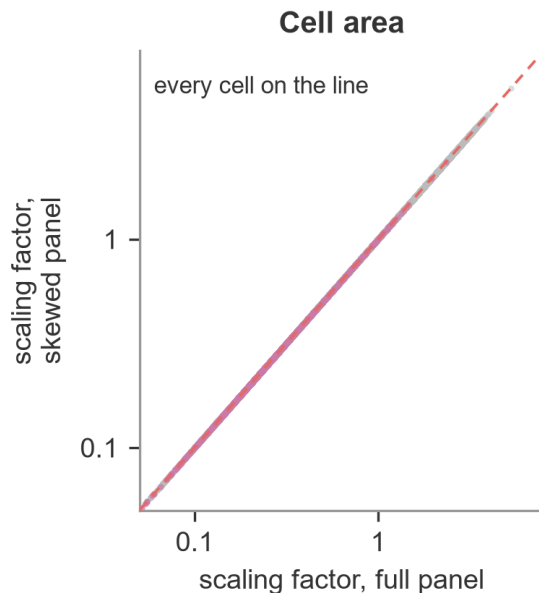
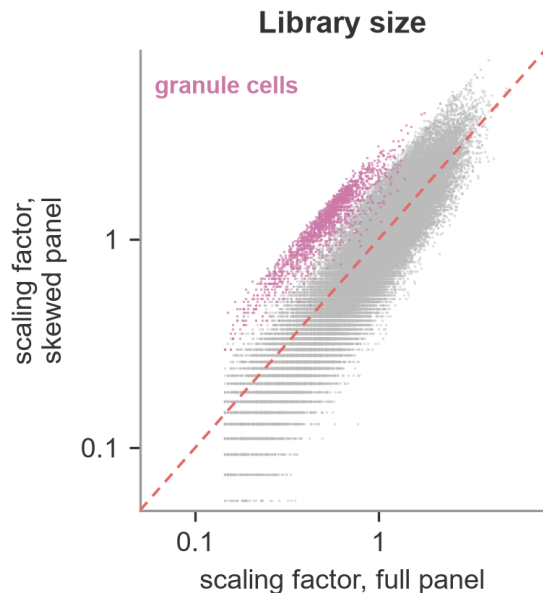


Same cell type, two regions



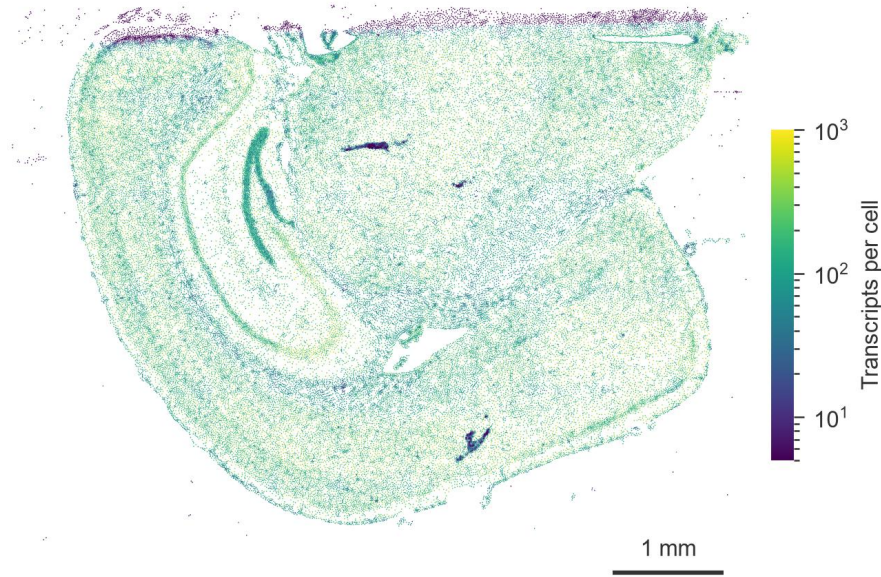
- 130 genes are higher in oligodendrocytes among neurons
- None are oligodendrocyte genes. *Slc17a7* (neurons): about 70x
- DE between regions shows the neighbours, not the cell
- Nucleus-only counts: 20 genes

Normalization: can we divide by total counts?

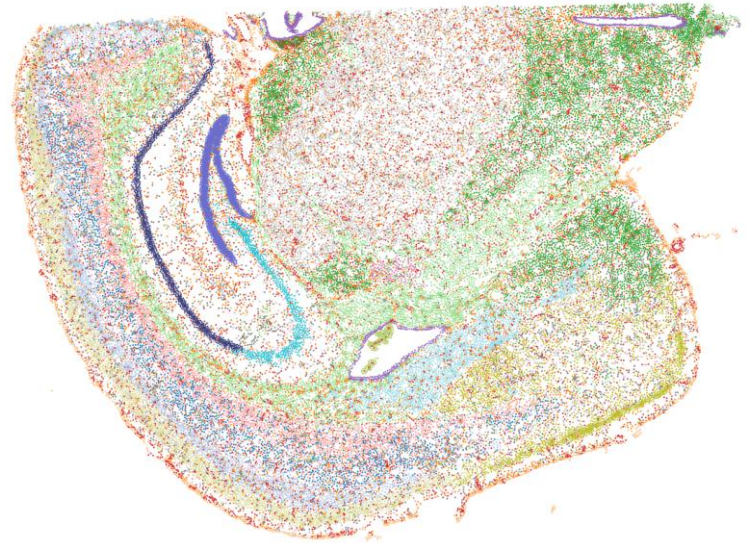


- scRNA-seq: divide by total counts (they measure sequencing depth)
- Here, the total depends on which genes are on the panel
- Cells the panel favours get large factors: their genes look too low
- Cell area does not depend on the panel

Total counts also follow the anatomy



normalize_total + log1p (24 Leiden clusters)



Ways to normalize

Method	Code	Watch out
Library size + log	<code>sc.pp.normalize_total; sc.pp.log1p</code>	depends on the panel; choose <code>target_sum</code>
Cell area + log	<code>X / cell_area</code>	depends on segmentation
Pearson residuals	<code>sc.experimental.pp.normalize_pearson_residuals</code>	dense matrix, memory
SCTransform	Seurat	uses total counts too
scVI	<code>scvi.model.SCVI</code>	needs raw counts in a layer

`target_sum` only sets the scale before the log. Our median is about 180 counts: `1e4` multiplies by about 55.

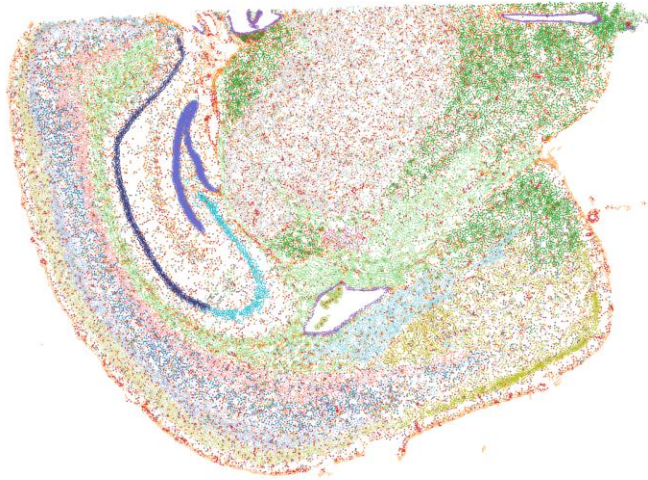
Which normalization?



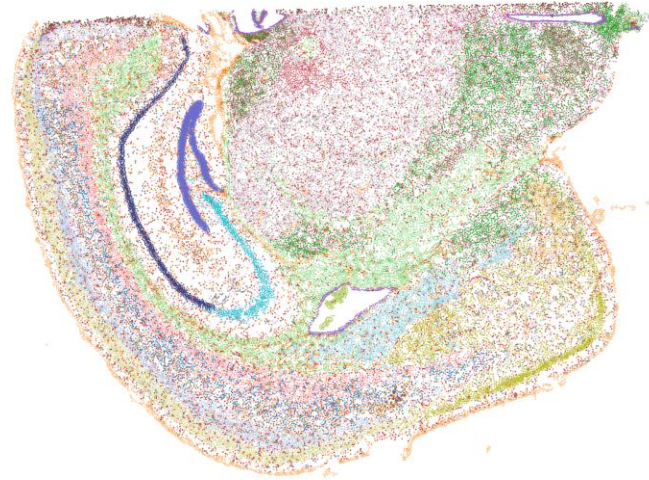
No independent benchmark yet for 5K or whole-transcriptome panels.

Normalization changes the clusters

normalize_total + log1p (24 Leiden clusters)



Cell area + log1p (28 Leiden clusters)



1 mm

Library size vs cell area: ARI = 0.70 (1 = identical clusters)

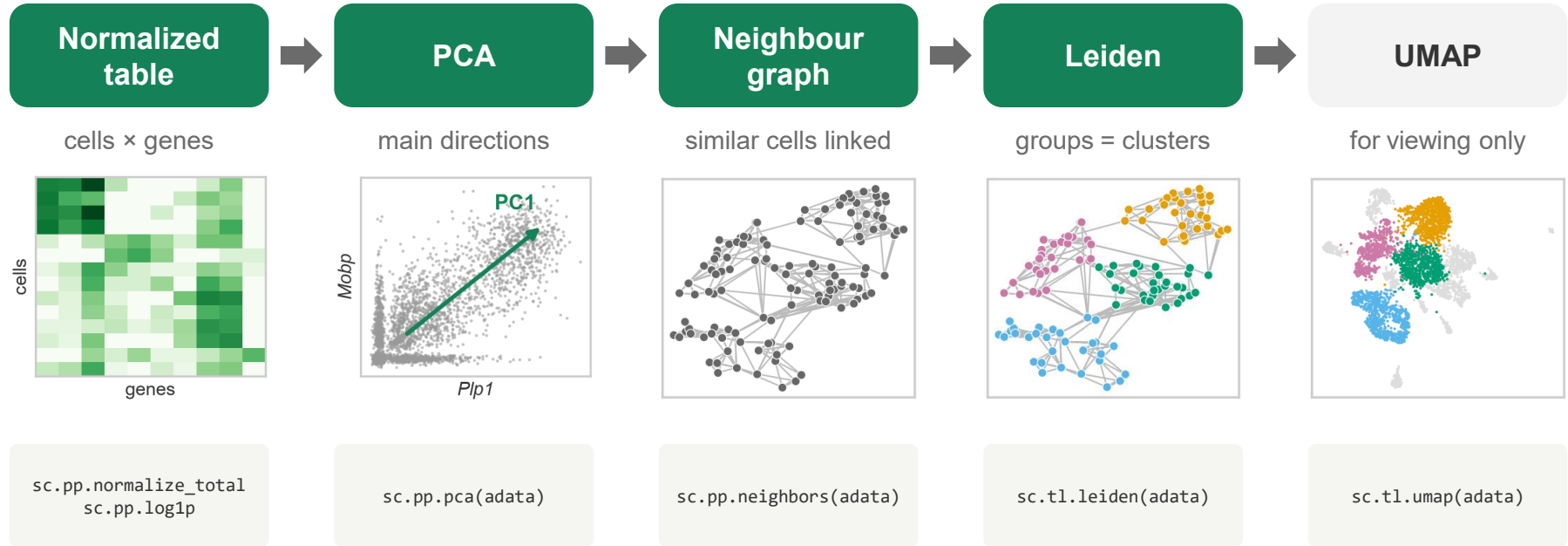
Small panels are different

	Targeted panel (100-1,000 genes)	Large panel (5K, WTX)
Gene selection (HVG)	usually skip	yes
Principal components	about 30	30-50
Counts per cell	tens to hundreds	hundreds to thousands
Normalization	library size or area	library size, Pearson, scVI

Always keep raw counts: `adata.layers["counts"] = adata.X.copy()`

From table to clusters

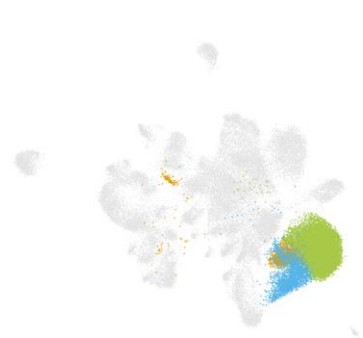
Unsupervised clustering: no labels are given (unlike **classification**, which learns from labelled cells).



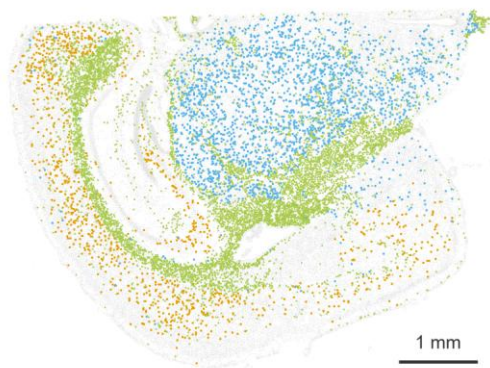
Clusters only show what the panel can see

Three oligodendrocyte clusters

- white matter: 11% neuron transcripts
- cortex: 30% neuron transcripts
- thalamus: 23% neuron transcripts



UMAP



- No marker on the panel, no cluster
- Neighbours can split a cell type: three oligodendrocyte clusters
- Check each cluster in space and with its markers
- Name clusters by markers, not by number

Summary

- 1 Read the run summary. Look at QC values on the tissue
- 2 Remove or flag bad regions first
- 3 Lenient cell filter, scaled to the panel. Check what you remove
- 4 Normalize: library size or area, keep raw counts
- 5 All genes, about 30 PCs, Leiden. Check clusters in space

Worth trying

MAD or combined QC scores

Area normalization

target_sum = 100
(Xenium, one study)

Contamination-aware segmentation

Practical 2

- 1 Explore the Xenium table: counts, areas, controls
- 2 Plot QC values on the tissue
- 3 Choose cut-offs and plot what you remove
- 4 Compare three normalizations:
library size, Pearson residuals, cell area
- 5 Cluster and find marker genes

Notebook

`p2_imaging_based_data_analysis.ipynb`